The reduction of OEIS A396807 modulo 10

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Abstract

OEIS A396807 is defined by a functional equation for its ordinary generating function and carries a congruence conjectures of P. D. Hanna. It is true. The equation determines the coefficients one at a time, with $a(N)$ entering with coefficient 1, so it determines them modulo 10 as well; and modulo 10 the series $x/(1-x)$ satisfies the equation exactly, because its compositional iterates are $x/(1-jx)$ and the relevant product of denominators collapses. Uniqueness then identifies the reduction, which is the conjectures. The whole argument is an identity between rational functions over $\mathbb{Z}/10\mathbb{Z}$; nothing is truncated or estimated.

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1 The sequence and the conjecture

OEIS A396807 is "G.f. satisfies A(x) = x + A^5(x)*A^6(x).". It has offset 1 and begins

$$1, 1, 11, 201, 4721, 129671, \ \ldots$$

Writing $A(x) = \sum_{n\geq 1} a(n)x^n$ and A^j for the j-th compositional iterate of A, the entry defines A by

$$A(x) \ = \ x + A^5(x)\, A^6(x). \tag{1}$$

This is the only input taken from the entry.

The entry separately carries the following comments:

> Conjecture: a(n) = 1 (mod 10) for n >= 1.
>
> Conjecture: [x^n] A^k(x) == k^(n-1) (mod 10) for n >= 1 and all integer k. — *Paul D. Hanna*

As of the "Last modified" line on the live entry (Jun 18 2026 00:43:54, revision 9) they are still recorded as conjectures, and nothing on the entry records them as settled either way.

2 The equation determines the sequence

Lemma 1. *Let $A(x) = \sum_{n\geq 1} a(n)x^n$ with $a(1) = 1$. Then for every $j \geq 1$ and every $N \geq 1$,*

$$\left[x^N\right] A^j = j\, a(N) + \big(\textit{a polynomial in } a(1), \ldots, a(N-1)\big).$$

Proof. Induction on j. For $j = 1$ the statement is $\left[x^N\right] A = a(N)$. Suppose it holds for j, and write $B = A^j$, so $\left[x^1\right] B = 1$ and $\left[x^N\right] B = j\, a(N) + \cdots$. Then $A^{j+1} = A \circ B = \sum_{m\geq 1} a(m) B^m$ and

$$\left[x^N\right] A^{j+1} = \sum_{m=1}^{N} a(m)\, \left[x^N\right] B^m.$$

The term $m = N$ contributes $a(N)\big([x^1]B\big)^N = a(N)$. The term $m = 1$ contributes $[x^N]B = j\,a(N) + \cdots$. For $2 \le m \le N-1$, every monomial of $[x^N]B^m$ is a product of $m \ge 2$ coefficients of B with indices summing to N, hence each index is at most $N-1$, so no $a(N)$ occurs. Adding up gives $(j+1)a(N)$ plus terms in $a(1), \ldots, a(N-1)$. $\square$

Lemma 2. *There is exactly one sequence of integers $a(1), a(2), \ldots$ satisfying (1), and for each $N \ge 2$ the equation read at order x^N has the form $a(N) = ($a polynomial with integer coefficients in $a(1), \ldots, a(N-1))$.*

Proof. Comparing coefficients of x in (1) gives $a(1) = 1$. Let $N \ge 2$ and collect the coefficient of $a(N)$ on each side of (1) at order x^N, using Lemma 1. On the left the coefficient is 1. On the right, x contributes nothing and the product A^5A^6 contributes nothing either: every monomial of $[x^N]\big(A^5A^6\big)$ is a product of one coefficient of A^5 and one of A^6 with indices $i, j \ge 1$ and $i + j = N$, so both indices are at most $N-1$. Hence the net coefficient of $a(N)$ is 1. Since that coefficient is 1, the order-x^N equation solves for $a(N)$ over $\mathbb{Z}$, and induction gives existence and uniqueness. $\square$

Corollary 3. *Let $m \ge 2$. If $\bar{A} \in (\mathbb{Z}/m\mathbb{Z})[[x]]$ satisfies $\bar{A} = x + O(x^2)$ and (1) with all coefficients read modulo m, then $a(n) \equiv [x^n]\bar{A} \pmod{m}$ for every n.*

Proof. Reduction modulo m is a ring homomorphism, so it carries the integer recursion of Lemma 2 to the same recursion over $\mathbb{Z}/m\mathbb{Z}$, where the coefficient of $a(N)$ is still 1 and hence still invertible. That recursion therefore has a unique solution, and both $a \bmod m$ and the coefficient sequence of $\bar{A}$ solve it. $\square$

3 The reduction modulo 10

Write $f(x) = \dfrac{x}{1-x}$ over $\mathbb{Z}/10\mathbb{Z}$.

Lemma 4. $f^j(x) = \dfrac{x}{1-jx}$ *for every $j \ge 1$, and more generally for every integer j, where f^{-1} denotes the compositional inverse.*

Proof. For $j = 1$ this is the definition. If $f^j(x) = x/(1-jx)$ then

$$f^{j+1}(x) = f\big(f^j(x)\big) = \frac{\dfrac{x}{1-jx}}{1 - \dfrac{x}{1-jx}} = \frac{x}{(1-jx) - x} = \frac{x}{1-(j+1)x}.$$

The maps $x \mapsto x/(1-jx)$ therefore form a group isomorphic to $(\mathbb{Z}, +)$ under composition, which also gives the statement for $j \le 0$; in particular $f^{-1}(x) = x/(1+x)$. $\square$

Lemma 5. *Over $\mathbb{Z}/10\mathbb{Z}$, f satisfies $f = x + f^5f^6$.*

Proof. By Lemma 4, $f^5f^6 = \dfrac{x^2}{(1-5x)\big(1-6x\big)}$, while $f - x = \dfrac{x}{1-x} - x = \dfrac{x^2}{1-x}$. So the claim is that $(1-5x)(1-6x)$ and $1-x$ agree in $(\mathbb{Z}/10\mathbb{Z})[x]$. Expanding,

$$(1-5x)\big(1-6x\big) = 1 - 11x + 30x^2,$$

and modulo 10 we have $11 \equiv 1$ and $30 \equiv 0$. Hence both sides equal $1-x$, and the two rational functions agree. $\square$

Theorem 6. *Over $\mathbb{Z}/10\mathbb{Z}$ the reduction of A is $x/(1-x)$. Equivalently $a(n) \equiv 1 \pmod{10}$ for every $n \geq 1$, and more generally*

$$\big[x^n\big] A^{j}(x) \;\equiv\; j^{\,n-1} \pmod{10}$$

for every $n \geq 1$ and every integer j. This is the conjectures of Section 1.

Proof. By Lemma 5, $f = x/(1-x)$ satisfies the reduced equation, and $f = x + O(x^2)$. By Corollary 3 the reduction of A equals f, so $a(n) \equiv \big[x^n\big] f = 1$.

Reduction modulo 10 carries composition of power series with unit linear coefficient to composition, so A^j reduces to f^j for every integer j — for $j < 0$ because A has linear coefficient 1, hence is invertible under composition, and the reduction map is a homomorphism of the composition group. By Lemma 4, $f^{j}(x) = x/(1-jx) = \sum_{n\geq 1} j^{\,n-1} x^n$, which gives the second statement. □

Remark 7. The proof used only that $(1-5x)(1-6x) \equiv 1-x$ modulo 10, that is, that 10 divides both $2 \cdot 5$ and $5 \cdot 6$. The largest modulus with that property is $\gcd\big(2p,\, p(p+1)\big)$, which is $2p$ for odd p and p for even p; here $p = 5$ gives 10. This is exactly why the statement is made modulo 10 and not modulo a larger number.

4 Verification

Three checks, each independent of the algebra above.

First, the recursion of Lemma 2 was run in exact rational arithmetic and its output compared against the entry's DATA at every one of the 19 published indices; the values agree exactly, and every one is an integer. A recursion that reproduced only some of them would mean the functional equation had been transcribed wrongly.

Second, at every step the coefficient of $a(N)$ was computed rather than assumed, by evaluating the order-x^N equation at $a(N) = 0$ and at $a(N) = 1$ and subtracting; it came out 1 at every N up to 24, as Lemma 2 requires. The same run was continued to $n = 24$ and every term reduced modulo 10; all residues are 1.

Third, the composition of the proof was carried out independently as a formal power series over $\mathbb{Z}/10\mathbb{Z}$ to order 30, starting from the coefficients of $x/(1-x)$ rather than from the closed form; both sides of (1) agree at every order, and the iterates f^{j} were confirmed to have coefficients $j^{\,n-1}$ for $j \leq 6$.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A396807.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 2, Cambridge University Press, 1999.

[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.